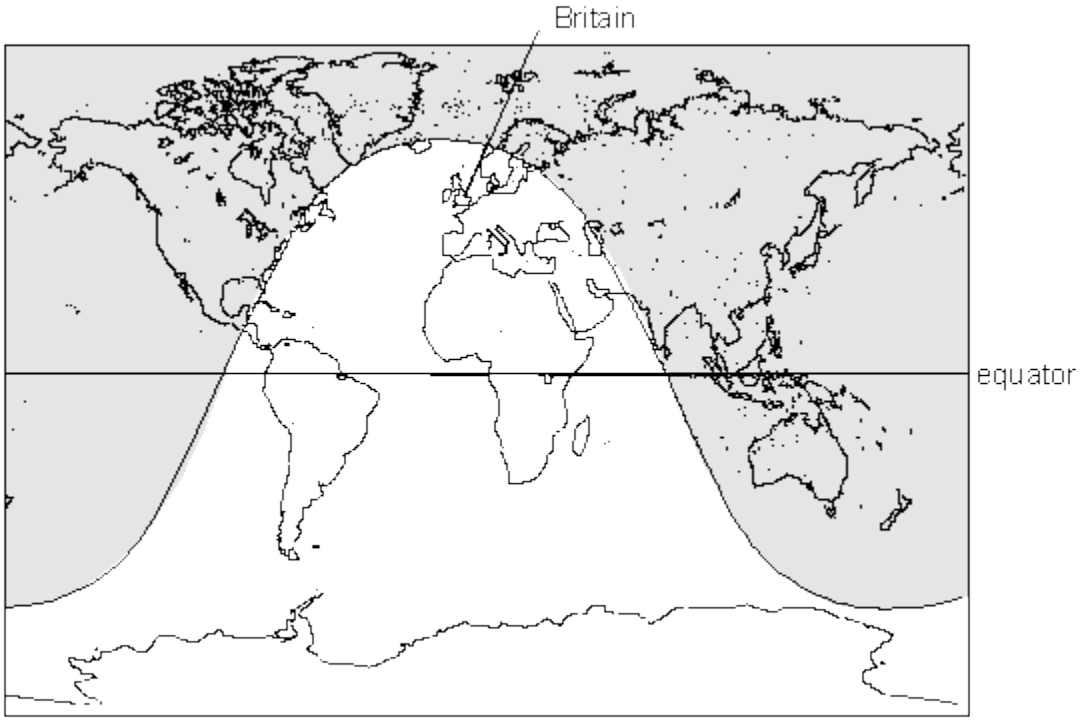
1. (a) The diagram shows a map of the world.



The shaded areas show where it is night-time at one moment in the year.

(i) At the time shown on the map, what season is it in Britain?
Give a reason for your answer.

.....

.....

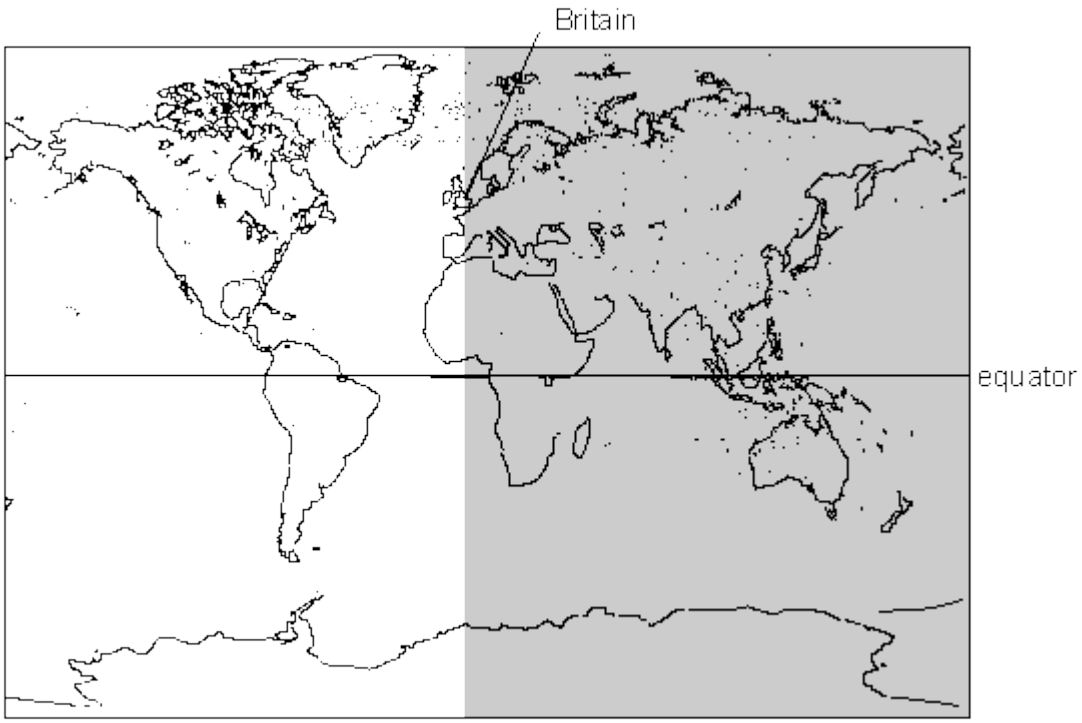
1 mark

(ii) At the time shown on the map, approximately what is the time in Britain?

.....

1 mark

(b) The map below shows a different pattern of night and day.



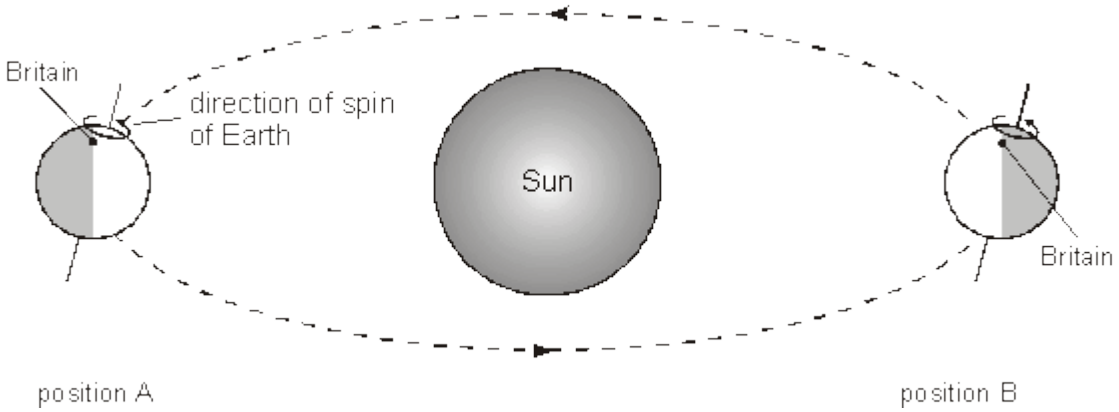
The pattern of night and day shown on the map occurs in only **two** months of the year. Which months are these?

1. .....

2. .....

2 marks
Maximum 4 marks

2. The diagram shows the Earth's orbit. The Earth is shown in two positions six months apart. In one of the positions it is midsummer in Britain, and in the other positions it is midwinter.



(a) Estimate what time of the day it is in Britain when the Earth is in position B, and explain your answer.

.....

.....

1 mark

(b) The Earth has rotated **exactly** 183 times on its axis as it moved between the two positions shown.

(i) Explain how you can tell from the diagram that the Earth has rotated a whole number of times.

.....

.....

1 mark

(ii) Explain how you can tell from the diagram that the time of day in Britain is different in the two positions.

.....

.....

1 mark

(iii) While the Earth has rotated exactly 183 times, only 182.5 days have passed. Calculate the exact time taken for each rotation of the Earth on its axis.

.....

.....

1 mark

(c) The Moon orbits the Earth approximately once each month. However, only one face of the Moon is ever visible from the Earth.

(i) Explain why we always see the same face of the Moon.

.....

.....

1 mark

(ii) Suggest how long in Earth days the period from sunrise to sunrise will be at a point on the Moon's surface.

.....

1 mark

Maximum 6 marks

Mark schemes

1.

(a) (i) **both** the season and the reason are required for the mark

any **one** from

- winter, because it is dark at the North Pole
- winter, because the nights are longer than the days

accept 'winter, because the nights are long'
***or** 'winter, because the days are short'*
accept 'winter, because most of the Northern Hemisphere is dark'
***or** 'winter, because the North Pole is tilted away from the Sun'*

1

(ii) noon **or** midday **or** 12.00

accept answers between 11.00 and 13.00

1

(b) **Answers may be in either order**

March

1

September

1

[4]

2.

Both the time and the explanation are required for the mark

(a) a time between 3.30 pm and 6.00 pm or later afternoon **or** sunset **or** dusk

accept any time between 15.30 and 18.00 hours

because Britain is moving from the day side of the world to the night side

accept as explanations 'because Britain is going from day to night or from light into darkness'
***or** 'because it is nearly night-time'*
***or** 'because the Sun is in the West'*
accept 'because it is sunset or dusk' as an explanation provided that it is not also given as the time.

1

(b) (i) Britain is in the same position in both diagrams

*do **not** accept 'Britain is in the same position relative to the Sun **or** to the sunlight' **or** 'from the position of Britain'*

1

(ii) it is dusk in one position and dawn in the other

accept 'it is morning in one diagram and evening in the other'
or *'the Sun is shining from different sides'*

accept 'in position A it is dawn' ***or*** *any time between 3.30 am and 6.00 am*

do ***not*** *accept a restatement that it is dusk in position B*

do ***not*** *accept 'it is light in one position and dark in the other'*
or *'Britain is spinning towards the Sun in A and away from it in B'*

1

(iii) 0.997 days

the unit is required for the mark

accept more decimal places
accept '182.5 × 24 ÷ 183 hours'
accept '23.9 hours' ***or*** *'23 hours 56 minutes'*

1

(c) (i) the time for the Moon to rotate on its axis is equal to the period of its orbit about the Earth

accept 'it turns on its axis once during a complete orbit of the Earth'

1

(ii) 27.3 days

accept '27 ***or*** *28 days'*

1

[6]